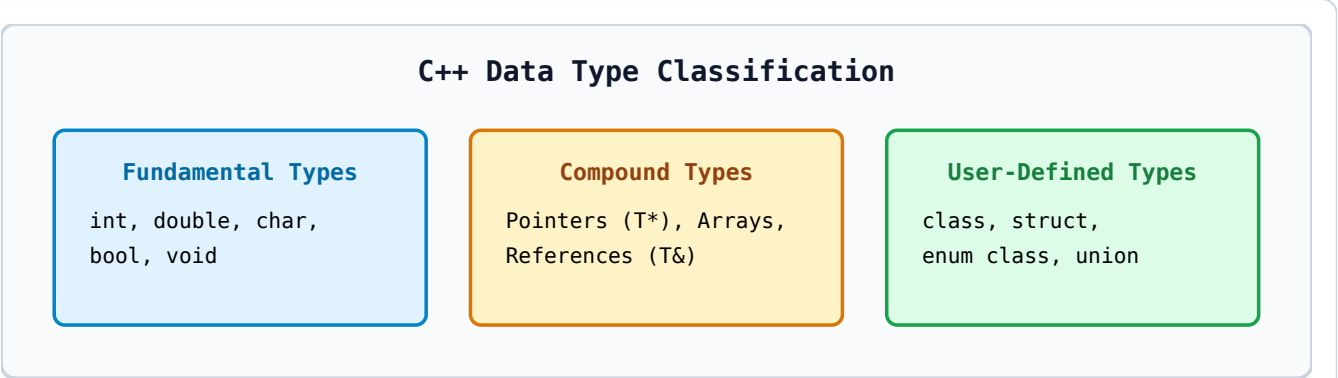
02 DATA TYPES AND VARIABLES • THEORY

01 fundamental types and representations

Detailed Theory, Exact C++ Rules, Memory Models & Exam Reference

01 — Fundamental Data Types and Two's Complement Representation



Data Type Hierarchy

1. Standard Minimum Bit Widths (`[basic.fundamental]`)

- `bool` : 1 byte (`true` or `false`).
- `char` : Guaranteed `sizeof(char) == 1` byte (at least 8 bits). Note: Whether plain `char` is signed or unsigned is **implementation-defined**!
- `short` : At least 16 bits (typically 2 bytes).
- `int` : At least 16 bits (**4 bytes / 32 bits** on all modern 32/64-bit desktop/server platforms, range -2^{31} to $2^{31}-1$).
- `long long` : At least 64 bits (8 bytes).
- `double` : 64-bit IEEE-754 floating-point (~15–17 decimal digits of precision).